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Source: Freshwater Science, 32(1):250-266.

Published By: The Society for Freshwater Science

DOI: <http://dx.doi.org/10.1899/12-008.1>

URL: <http://www.bioone.org/doi/full/10.1899/12-008.1>

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Survival of aquatic invertebrates in dry bed sediments of intermittent streams: temperature tolerances and implications for riparian management

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Abstract. Physicochemical conditions in refugia must remain within the tolerance limits of aquatic invertebrates if they are to persist in intermittent streams when surface water is absent. We investigated shallow streambed sediments as a refuge for aquatic invertebrates and survival of invertebrates at high temperatures during the dry period in 3 headwater intermittent streams in New Zealand. Sediments from dry stream beds inundated in the laboratory contained 37 of the 53 taxa found in nearby flowing streams, a result suggesting that many taxa attempt to use the dry stream bed as a refuge during dry periods. Relative abundances of microcrustacea (copepods and ostracods), Gastropoda and Coleoptera (mostly Scirtidae) were greater in rewetted sediments than in sediments from nearby flowing streams, and several Ephemeroptera, Plecoptera, and Trichoptera taxa were present in low densities in rewetted sediments. All taxa that became active in rewetted sediments, except Hydrobiosidae, *Tarapsyche olis* (Trichoptera), *Lymnaea* (Gastropoda), Chironomini, and Cladocera, were active within a few hours of rewetting. We tested invertebrate survival in rewetted sediments at high temperatures (comparable to near-surface temperatures of dry stream beds in the field) in 2 experiments. In experiment 1, abundances of all major taxa were 11 to 94% lower in warm (air temperature = 30°C, sediment temperatures = 22–25°C for 1 h/d) than in cooler treatments. In experiment 2, abundances of most major taxa except Coleoptera and Acari were >75% lower in warm (air temperature = 37°C, sediment temperature = 32°C for 1 h/d) than in cooler treatments. Mid-summer daily maximum temperatures in the top 5 cm of the stream bed were 35 to 42°C in open pasture, 17 to 19°C under dense forest canopy, and 22 to 29°C under sparse riparian trees. Results suggest that maintaining the microclimate provided by riparian forest is important for protecting the aquatic invertebrate community in intermittent streams.

Key words: intermittent streams, refugia, benthic invertebrates, temperature tolerance, drought, riparian management.

Aquatic invertebrates use various combinations of life-history strategies and dry-season refugia to maintain long-term populations in intermittent streams. A dry-season refuge is a habitat that has physicochemical conditions that enable aquatic taxa to persist or survive dormancy in one of their life-history stages throughout the dry period. Common dry-season refugia include in-channel permanent pools, leaf packs, woody debris or algae, shallow streambed sediments, deep bed sediments (hyporheic zone), riparian vegetation, and perennial reaches (Williams 2006, Robson et al. 2011). The types and physicochemical characteristics of refugia available at a particular site influence aquatic invertebrate recolonization dynamics, community succession (Paltridge

et al. 1997, Lake 2003, Fowler 2004, Chester and Robson 2011), and resilience with respect to length and intensity of the dry period (Williams 1996, Lake 2003, Larned et al. 2007). Species composition differs among streams with different characteristics, such as connectivity to permanent waters (Fritz and Dodds 2005, Sheldon et al. 2010), type and complexity of streambed substrates (Boulton 1989), and connectivity to the hyporheic zone (Fritz and Dodds 2004) as a consequence of refuge availability among other factors. Thus, identifying the refugia used by different taxa is fundamental to understanding the ecology and to protecting the fauna of intermittent streams (Bond et al. 2008, Datry et al. 2011).

Several authors have inferred the importance of specific refugia to stream recolonization after flow resumes (Fritz and Dodds 2004, Churchel and Batzer 2006), but few have directly examined the potential of

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dry streambed sediments as a refuge by rewetting dry sediment samples under controlled conditions. Those investigators who did so (Boulton 1989, Paltridge et al. 1997, Larned et al. 2007, Stubbington et al. 2009, Chester and Robson 2011, Datry et al. 2012) found that significant numbers of taxa remained viable in dry streambed sediments. However, the importance of surface sediments relative to other refugia varies among studies. In a sand-bed river in northern Australia, dry streambed sediments harbored more taxa than sediments in billabongs (permanent pools) and hyporheic sediments (Paltridge et al. 1997), whereas in 2 other studies in Australia (Boulton 1989, Chester and Robson 2011) and 1 in Canada (Williams and Hynes 1977), refugia without free water contained fewer taxa than remnant pools.

The relative importance of different dry-season refugia might vary among stream and river types. The suitability of potential refugia depends on their physicochemical characteristics, which may be affected by factors such as dry-season length and intensity, streambed substrate type, depth to water table, presence of upstream perennial reaches, and type of riparian vegetation. These factors also influence the composition of aquatic invertebrates requiring refuge. For example, in a New Zealand gravel-bed river, invertebrate richness in dry bed sediments decreased with increasing duration of the dry period (Larned et al. 2007). Lentic conditions, extreme O_2 , temperature, pH, and conductivity can render remnant pools unsuitable for some taxa (Linklater 1994, Williams 2006). Differences in physical habitat, flow and drought dynamics, and biological communities among sites may explain the conflicting evidence regarding the hyporheic refuge hypothesis (Dole-Olivier 2011, Stubbington 2012). Therefore, examining the relative importance of potential refugia in different environments is necessary to develop conceptual models of intermittent stream ecology that are transferable across stream types (Datry et al. 2011). Our first goal was to assess the importance of surface and shallow subsurface streambed sediments as dry-season refugia for invertebrate taxa in intermittent headwater streams on the east coast of New Zealand's North Island. We expected dense forest cover, relatively mild summer temperatures, coarse gravel beds, and shallow water tables at our native forested sites to allow a wider range of taxa to take refuge in the bed sediments of these streams than of those in some previous studies.

The rate at which invertebrates become active after water returns to a dry stream bed has significant consequences for survival and competition in an unpredictable environment. Responding too quickly

could lead to mass mortality if an isolated summer rainfall causes a false start to the flow period (Bishop 1967, Brendonck and De Meester 2003), whereas responding too slowly may mean loss of a competitive advantage or insufficient time to complete development (Williams 2006). However, the speed with which taxa undergoing full diapause or over-summering as eggs can respond to rewetting may be limited (Williams 2006). Therefore, our 2nd goal was to record the rate at which invertebrate taxa in dry bed sediments responded to rewetting. We expected rates to differ among taxa depending on life-history stage used to survive dry periods, and that some taxa may exhibit a delayed response that would protect individuals from false starts.

Several investigators have examined the habitats used as dry-season refugia by aquatic invertebrates, but very few have examined physicochemical aspects of these habitats as refugia. Water temperature affects species survival in perennial and intermittent waters (Quinn 2000, Williams 2006), but we know of only 1 study in which the effect of high sediment temperatures on invertebrate survival in dry stream beds was examined. This case was an extreme example in which bed temperatures were $>60^\circ\text{C}$ (Boulton et al. 1992). Therefore, our 3rd goal was to estimate the temperature limits within which dry streambed sediments remain viable refugia for over-summering aquatic invertebrates. We expected these limits to be higher than the temperature limits of active larvae because some invertebrates use resting stages as a strategy for surviving high temperatures (Pritchard et al. 1996, Brendonck and De Meester 2003, Williams 2006). However, results of a few studies (Brittain 1977, Friesen et al. 1979, Juliano et al. 2002) suggest that the difference in thermal tolerances between active and resting stages may be small.

Awareness of biodiversity and ecological processes in intermittent streams is growing, but much less is known about management strategies required to maintain valuable aspects of these systems. Recent calls for protection of intermittent stream communities have focused mostly on maintaining the hydrological regime (Lake 2003, Williams 2006, Larned et al. 2010) or protecting remnant pools (Sedell et al. 1990, Sheldon et al. 2010). Investigators advocating protection of other refugia (Bond et al. 2008, Datry et al. 2012, Dole-Olivier 2011, Stubbington 2012) have focused on human impacts, such as recreation, sediment mining or deposition, hydrological disturbance, and pollution. However, removal of riparian vegetation is one of the most widespread human impacts on streams in New Zealand (Quinn 2009) and elsewhere (Brierley et al. 2005, Palmer et al. 2007), and

TABLE 2. Laboratory treatments (rewet without temperature treatment, 6-wk cool- or warm-temperature incubation before rewetting), sediment collection depths in the dry stream bed (0–5 or 5–10 cm), and number of replicate dry sediment samples (each 0.25 m²) collected at each site for use in laboratory rewetting experiments in 2006 and 2007. INT1–3 are sampling sites with different canopy cover (NF = native forest, P = open pasture, S = shaded pasture) on intermittent streams.

Collection date	Site	Rewet immediately		Cool or warm-temperature incubation before rewetting	
		0–5 cm	5–10 cm	0–5 cm	5–10 cm
February 2006	INT1 NF	4	4	4	
	INT1 P	4	4	4	
	INT1 S	4	4	4	
	INT2 NF	2	2	2	
	INT3 NF	2	2	2	
January 2007	INT1 NF			4	
	INT1 P			4	4
	INT2 NF			2	

sample locations. All of the study streams had similar substrate that was dominated by pebbles (4–64 mm) and cobbles (64–256 cm). However, at all INT1 sites the pebble/cobble bed was underlain by a mixture of sand and clay below a depth of ~10 cm, whereas INT2 and INT3 had an open pebble/cobble matrix to a depth of >30 cm. Visual inspection revealed that the more open streambed matrix at INT2 and INT3 allowed some subsurface flow after surface water had disappeared, whereas subsurface flow was negligible at INT1.

The primary vegetative cover in the study catchments was pasture for sheep and beef farming, but some pockets of native forest remained, mostly in river gullies. On each stream (INT1–3), we established a native forest site (NF) in a patch of intact native forest ≥30 m wide on each bank and extending ≥300 m upstream of the study site. Riparian vegetation cover directly over the stream channel provided ~80% shading (estimated by eye). On INT1, we established an open-pasture site (INT1 P) 50 m downstream of the native-forest remnant and a pasture site shaded by scattered trees that provided ~50% shading (INT1 S) ~300 m downstream of the forest remnant. Logistical constraints limited us to 1 P and 1 S site.

Benthic communities in flowing streams

We used a Surber sampler (250-μm-mesh net) to collect benthic macroinvertebrate samples from riffles at sites INT1–3 during flow periods. We collected 4 samples/site every 2 wk from the end of March 2005 (2 wk after flowing water appeared) until May to capture potentially rapid changes early in the season and once per month thereafter until November 2005 (Table 1). We characterized the benthic community at site INT1 P from Surber samples collected in a similar

manner once in August 2005. This August sample probably represented the typical community at this site because benthic community composition at the NF sites was reasonably stable between June and October (Storey and Quinn 2008), and the composition at both INT1 P and NF sites was similar to that found in subsequent years (RGS, unpublished data).

Aquatic invertebrates in dry sediments

We collected dry sediments from all sites in mid-February 2006. We use the term dry sediments to mean without free water, but the sediments remained moist (up to 15% moisture content by mass; see Results). We collected dry sediments in riffle areas where surface water was absent by removing the top 5 cm of streambed sediment from a 0.25-m² area with a trowel. We collected the next 5-cm layer of sediment separately to represent subsurface sediments. The number of 0.25-m² replicates collected from each site and depth was variable (Table 2) because the remoteness of sites INT2 and INT3 limited the number of samples we could collect and carry. We placed each ~12.5-L dry-sediment sample in a plastic bag and returned it to the laboratory where it was put in an 18-L (50 cm × 30 cm × 14 cm) plastic tub and inundated with dechlorinated tap water. We kept the tubs at a constant temperature of 15°C. We used an air bubbler in each tub to maintain dissolved O₂ concentrations in the overlying water close to saturation. After 0.5, 3, 8, 14, and 21 d of submersion, we sampled invertebrates by rocking the tubs back and forth to dislodge invertebrates from sediments and pouring the overlying water through a 125-μm-mesh net (this mesh size retained small animals, such as 1st-instar larvae). We returned the water to the tub and repeated the process once. We washed any large rocks in the tub separately into the net to collect attached

invertebrates. We used this method to balance thoroughness against the risk of damaging live animals, but we may have missed some animals. We transferred the invertebrates collected to a small tray, counted only living individuals, and identified them under 10× magnification to the lowest possible taxonomic level. We were able to identify most stonefly and some caddisfly individuals only to order and mayflies and hydrobiosid caddisflies only to family. We identified individuals in all other taxa to the same taxonomic level as in the flowing-stream benthic invertebrate survey. We based identifications on the keys of Winterbourn et al. (2000) for insects and Winterbourn (1973) for mollusks. We recorded the life stage or approximate size of mayflies, stoneflies, caddisflies, beetles, copepods, and snails in qualitative terms, e.g., hatching from egg/1st instar/small/medium/large/pupa. We returned living invertebrates to the tubs they had come from (following Boulton 1989 and Paltridge et al. 1997).

Sediment moisture content and streambed temperature

We measured sediment moisture content of 60-g subsamples of each sample after removing particles >4 mm diameter so that differences in moisture content were not confounded by differences in sediment particle size in these relatively small subsamples. We measured moisture content by weighing the subsamples before and after oven-drying them for 10 h at 105°C. We measured sediment moisture content both for ambient field conditions (2007; 2 replicates/sample sealed in plastic bags and measured immediately on return to the laboratory) and after exposure to the 2 temperature treatments (2006; 1 replicate/sample).

We measured field temperatures in dry stream beds by burying temperature loggers (StowAway TidbiT TBI32-05+37, Onset Computer Corporation, Cape Cod, Massachusetts) at depths of 0, 5, 10, and 15 cm below the streambed surface at INT2 NF, INT1 P, and INT1 S. Loggers recorded at 30-min intervals from 13 February to 3 April 2006, spanning the warmest and driest period of the summer.

Effects of sediment temperature on invertebrate survival

In 2006, we collected a second 0.25-m² sample from the top 5 cm of sediment adjacent to each sediment sample described above and split this 2nd sample into 2 equal parts (Table 2). We placed halves into 1 of 2 temperature-controlled incubators to measure the effect of temperature on survival of invertebrates in dry sediments. In the cool chamber, we set baseline temperature at 10°C, gradually increased the temper-

ature to 18°C each day between 0600 and 1200 h, and gradually decreased the temperature to 10°C between 1600 and 0100 h. A power outage on 1 occasion caused the temperature in the cool chamber to rise to 26°C for ~15 h during the treatment period. In the warm chamber, we set baseline temperature at 12°C, gradually increased it to 30°C, and gradually decreased it back to 12°C on the same schedule as described above. Based on measurements made during a similar experiment in 2007, we estimate that temperatures in the middle of the sediment samples in the warm chamber reached daily maxima of 22 to 25°C, similar to mid-summer temperatures in 5-cm-deep sediments at 50%-shaded sites and 15-cm-deep sediments at open pasture sites. After 6 wk, we removed the sediments from the chambers, inundated, and sampled them as described above after 1, 8, and 15 d of submersion (a lower frequency than in the 1st experiment because time trends were not the main focus, and the 1st experiment showed that almost all taxa were active within 3 d). For any taxon, we considered a large and significant difference in abundance between cool and warm treatments an indicator that the temperature tolerance of that taxon had been exceeded.

In late January 2007, we collected dry sediment samples to measure effects of the more extreme bed-surface temperatures. The dry period prior to sample collection was ~2 wk shorter in 2007 than in 2006, but streams dried more rapidly in 2007. The 2007 sampling design (Table 2) was focused on comparisons with INT1 P surface sediments, which were likely to experience the most extreme temperatures. Our purpose was to compare surface with subsurface sediments at INT1 P and to compare INT1 P with NF sediments. The cool chamber cycled daily between 10 and 18°C, and the warm chamber cycled daily between 20 and 37°C. Temperature loggers showed that temperatures in the middle of the sediment samples in the warm chamber reached daily maxima of 31 to 32°C for periods of ≤1 h, comparable to daily maximum temperatures in 5-to-10-cm-deep sediments at open pasture sites. After 6 wk in the chambers, we inundated sediment samples and sampled them as described above, recording only living individuals. Almost all invertebrate taxa were active within 3 d of rewetting in the 2006 experiment, so we collected samples for only 5 d after rewetting in 2007.

Data analyses

We removed semiaquatic taxa from the data set to keep the focus on fully aquatic taxa. When comparing

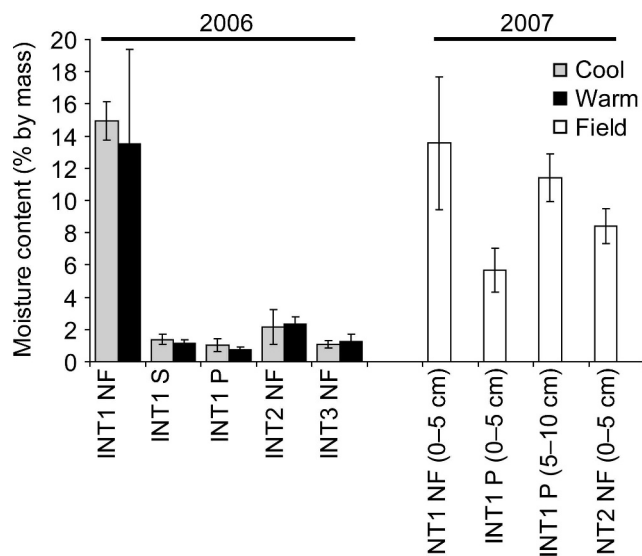


FIG. 1. Mean (± 1 SE) moisture content (% by mass) of streambed sediments after exposure to the cool and warm temperature treatments (2006 samples) and in the field (2007 samples). INT = intermittent stream, NF = native forest, S = shaded pasture, P = open pasture.

sites, depths, temperature treatments, and rewetted vs flowing-stream sediments, we defined abundance as the maximum abundance of each taxon observed during the first 8 d of rewetting in the 2006 experiments and the 5 d of rewetting in the 2007 experiments. Results of analyses done with the full 21 d and with the first 8 d of 2006 data did not differ significantly. Therefore, for the 2006 experiment, we used data from only the first 8 d after rewetting to maximize comparability with the 5-d period of the 2007 experiment. We used maximum abundance over the rewetted period instead of mean abundance because it was a better estimate of the true number of individuals that became active. The number of individuals of some taxa decreased in later samples, probably because individuals were overlooked or died. Either alternative would have led to underestimation of the true numbers that became active.

We compared invertebrate communities in rewetted and flowing-stream sediments with analysis of similarities (ANOSIM) on the Bray–Curtis similarity matrix of arcsin-transformed % abundance (PRIMER 6; PRIMER-E Ltd., Ivybridge, UK). We identified the taxa responsible for the differences between groups of samples with the similarity percentages (SIMPER) routine in PRIMER 6. We compared invertebrate abundance and richness with paired-samples *t*-tests, 1- and 2-factor analyses of variance (ANOVAs), and nonparametric equivalents (Mann–Whitney and Kruskal–Wallis tests) in SPSS (version 11; IBM SPSS,

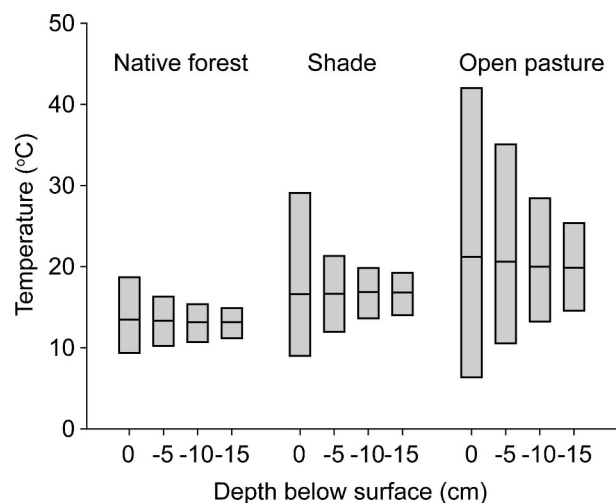


FIG. 2. Daily mean (lines in boxes), maximum, and minimum (ends of boxes) temperatures in dry stream beds during a 5-d period of settled sunny weather between 22 and 26 February 2006.

Armonk, New York). We measured changes in these variables over time with repeated measures ANOVA (Statistica, version 8; StatSoft, Tulsa, Oklahoma). We log($x + 1$)-transformed abundance data to improve normality and homogeneity of variance (tested by Levene's statistic). Taxon richness data needed no transformation.

Results

Sediment moisture content

Sediment moisture content of field samples in 2007 ranged from 6 to 14% by mass. Moisture content was higher in NF than in INT1 P surface samples ($t = 3.65$, $df = 22$, $p = 0.001$). Moisture content at INT1 P was higher in subsurface (5–10 cm) than in surface (0–5 cm) samples (Fig. 1). In the 2006 experiment, sediment moisture content after the temperature treatments ranged from 1 to 15% (Fig. 1) but did not differ between treatments (paired samples *t*-test, $t = 0.81$, $df = 13$, $p = 0.44$).

Daily temperature fluctuations in the field differed between NF, INT1 S, and INT1 P (Fig. 2). During a 5-d period of sunny weather from 22 to 26 February 2006, streambed surface temperatures at NF sites were $<19^{\circ}\text{C}$ at the sediment surface and $<17^{\circ}\text{C}$ at 5 cm below the surface. In contrast, surface temperatures at INT1 P exceeded the temperature logger limit of 38°C daily and reached $\sim 42^{\circ}\text{C}$ (estimated from the curve of the $\frac{1}{2}$ -h temperature readings), whereas daily temperatures 5 and 10 cm below the streambed surface reached 35 and 28°C , respectively. At INT1 S,

TABLE 3. Continued.

Phylum/order	Taxon	Flowing water					Rewetted dry sediments				
		INT1 NF	INT1 P	INT2 NF	INT3 NF	INT1 NF	INT1 S	INT1 P	INT2 NF	INT3 NF	
Cladocera	Cladocera	C	+	+			+	+		+	
Isopoda	Janiridae	+		+	+						
Amphipoda	Talitridae	+		+							
	<i>Paraleptamphopus</i>	+		+							
Gastropoda	<i>Potamopyrgus</i>	A	C	C	+	A	+	+	+	+	
	<i>Physa</i>	+	+			+	+	+			
	<i>Lymnaea</i>	+	+			+	+	+			
	<i>Gyraulus</i>	+	C		+	+	+	+	A	A	
Oligochaeta	Naididae	C	+	C	+	C	C	C	+		
	Enchytraeidae	+	+	+	+	+	+	+		+	
	Lumbricidae	+	+	+	+	+	+	C		+	
	Lumbriculidae	+	+	+		+	+	C		+	
Platyhelminthes		+	+				+	+	+	+	
Nematoda		C	A	+	+	C	A	C	C	C	
Hydra		+		+	+						
Eggs		C	+	C	C	C	C	+	C	+	
Total number of taxa		47	22	43	24	30	26	24	23	23	

temperatures were $<29^{\circ}\text{C}$ at the sediment surface and $<22^{\circ}\text{C}$ at 5 cm below the surface.

Aquatic invertebrates in rewetted compared to flowing-stream sediments

Based on the lowest taxonomic level at which animals in both the benthic survey and the rewetting experiment could be identified, 53 taxa occurred in flowing-stream sediment and 37 of those taxa were found in rewetted sediments (2006 and 2007 results combined; Table 3). Taxa not found in rewetted sediments included several Coleoptera (Hydraenidae, Ptilodactylidae, and the elmid *Hydora* sp.), several Diptera (Dixidae, *Austrosimulium* sp., Tabanidae, and the tanyderid *Mischoderus* sp.), isopods, and amphipods. In addition, several caddisfly genera (*Olinga*, *Helicopsyche*, *Oxyethira*, and *Oeconesus*) that were found in the flowing-stream sediments were probably not among the 4 unidentified caddisfly individuals found in the rewetted sediments.

All major taxa in the flowing-stream sediment were represented in the rewetted sediments (Fig. 3). However, the proportions of the major taxa differed significantly between invertebrate assemblages in rewetted and flowing-stream sediments (ANOSIM, $R = 0.419$, $p = 0.001$). Proportions of major taxa did not differ significantly among rewetted sediments from 2006, 2006 (cool), and 2007 (cool) (ANOSIM, 2006 vs 2006 cool: $R = 0.16$, $p = 0.11$; 2006 cool vs 2007 cool: $R = 0.272$, $p = 0.07$; 2006 and 2007 cool: $R = 0.374$, $p = 0.07$), even though 2006 (cool) and 2007 (cool) samples underwent an additional 6 wk of dry conditions relative to 2006 samples.

In the 2006 experiment, rewetted sediments differed from flowing stream samples largely because of lower proportions of Oligochaeta, Chironomidae, and Trichoptera, and higher proportions of microcrustacea (mostly Copepoda and Ostracoda), Gastropoda and Coleoptera (mostly Scirtidae), regardless of whether we used data from the first 8 or all 21 d. The higher proportion of microcrustacea in rewetted sediment samples might have been the result of the smaller mesh size of the net used to sample the rewetted sediments, but other taxa, which have larger body sizes, should have been captured with equal efficiency by both nets. Differences in Oligochaeta, Chironomidae, and Trichoptera between rewetted and flowing-stream sediments remained significant when microcrustacea were removed from the analysis.

Temporal patterns

In 2006, most fauna found in rewetted sediments, including members of all major taxa except caddisflies,

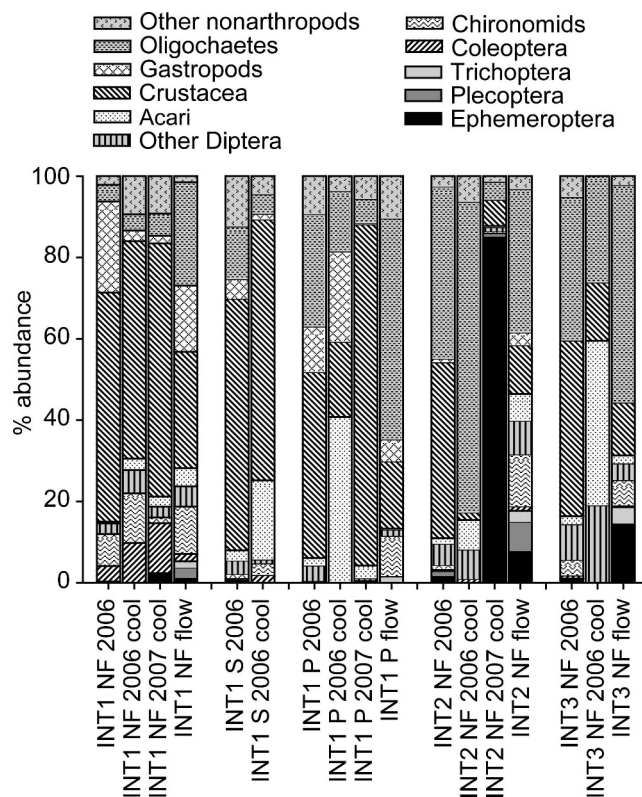


FIG. 3. Relative abundance of the major invertebrate taxa in rewetted streambed communities in 2006, the 2006 cool-temperature (10–18°C) treatment, the 2007 cool-temperature (10–18°C) treatment, and in benthic surveys of the flowing streams in 2005 (flow). INT = intermittent stream, NF = native forest, S = shaded pasture, P = open pasture.

became active within the first few hours of rewetting (Fig. 4). The only taxa not found on day 0 were Hydrobiosidae and *T. olis* (McFarlane) (Trichoptera), *Zelandotipula* sp. (Tipulidae), Chironomini, and Cladocera, which appeared on days 8, 3, 3, 14, and 21, respectively. The total number of invertebrates increased 62% between days 0 and 21. The overall increase in $\log_{10}(\text{abundance})$ was significant (repeated measures ANOVA, $F_{4,120} = 8.8$, $p < 0.001$), and significant increases occurred between days 8 and 14 and between days 14 and 21 ($F = 10.4$, $p = 0.003$ and $F = 7.0$, $p = 0.01$, respectively). Total abundance increased 54% between days 0 and 3 (Fig. 4), but this increase was caused mainly by extremely high numbers of ostracods, copepods, and *Potamopyrgus* (Gastropoda) in 1 tub on day 3 that did not appear in subsequent samples. $\log_{10}$ -transformation of total abundance reduced the influence of this single very high value on mean abundance, leaving a more gradual temporal trend (y_2 axis in Fig. 4). A few mayflies and stoneflies were active beginning on day 0 and throughout the experiment, whereas caddisflies

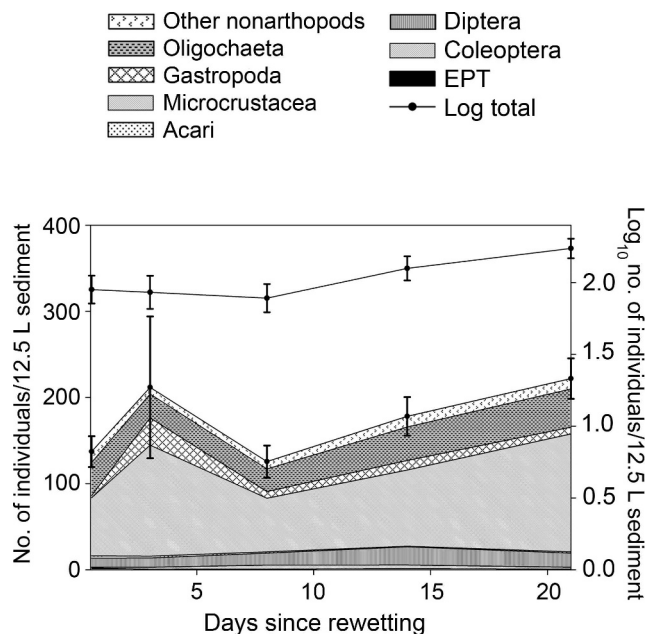


FIG. 4. Mean (± 1 SE) abundance and $\log_{10}(\text{total abundance})$ of invertebrate taxa in rewetted streambed sediments vs time after rewetting, averaged across all sites, depths, and replicates (data from 2006). EPT = Ephemeroptera, Plecoptera, and Trichoptera, INT = intermittent stream, NF = native forest, S = shaded pasture, P = open pasture.

were seen first on day 3, but not after day 8 (alive or dead). Numbers of gastropods, microcrustaceans, and oligochaetes were variable and had no overall trend between days 0 and 21. Taxon richness in all replicates combined was nearly constant (day 0: 28 taxa, day 21: 27 [data not shown]). Richness in sediments from individual sites (all replicates combined) changed from -15 to 25% relative to day 0 over the 21-d period.

In 2007, aquatic invertebrates collected from sediments held for 6 wk at cool temperatures before rewetting followed a temporal pattern similar to that described above (data not shown). Except for *Tarapsyche olis*, Chironomini, and *Lymnaea* sp. (Gastropoda), every taxon active on day 5 was active on day 0. Richness in sediments from each site (all replicates combined) changed by ≤ 1 taxon between days 0 and 5. Total abundance ranged from 200 to 1450/12.5 L sediment (i.e., /0.25 m² of streambed sampled) on day 0. Abundance increased 94% from day 0 to 5 in INT2 NF because of the appearance of many mayfly larvae and 128% in INT1 P because of the appearance of many ostracods, but rose only 27% in INT1 NF over the same period.

Life stages in rewetted sediments

All mayflies and most stoneflies, caddisflies, and scirtid beetles were assessed to be 1st-instar larvae, a result suggesting they may have hatched from eggs

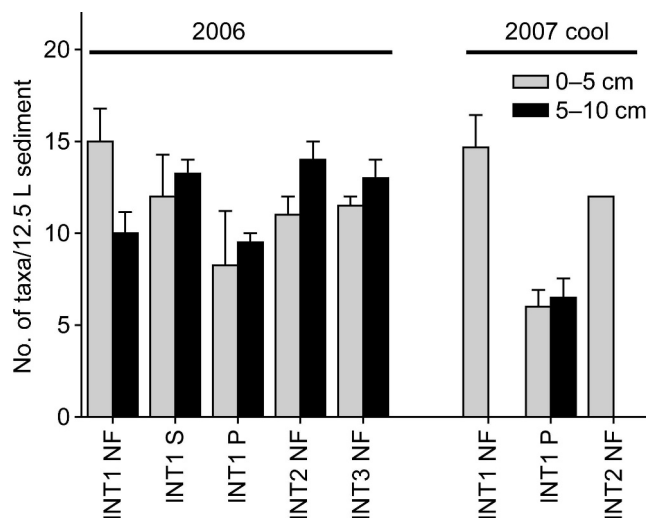


FIG. 5. Mean (± 1 SE) taxon richness (total number of taxa/12.5 L sample observed over the sampling period) in shallow (0–5 cm) and deep (5–10 cm) rewetted sediments at sites with different riparian vegetation types (NF = native forest, S = shaded pasture, P = open pasture). INT = intermittent stream.

after the sediments were inundated. One stonefly was found in the process of hatching. The other stonefly, caddisfly, and scirtid larvae were only slightly larger, and probably were not more than 2nd instars. Many harpacticoid copepods were found mating or carrying eggs, indicating they occurred as mature stages in the dry sediments. Gastropods (*Potamopyrgus antipodarum* (Gray), *Physa* sp., *Lymnaea* sp., and *Gyraulus* sp.) occurred in sizes from very small to occasionally large. Some Diptera pupae were either unidentified or belonged to Psychodidae or Chironomidae. Eggs probably were not sampled consistently because they are small relative to the net mesh, but we found up to 150 eggs/12.5 L sediment (mean $\pm$ SE, 7.1 ± 1.4) in 2006. Numbers decreased from a mean of 14 on day 1 to 3 on day 21.

Differences between riparian vegetation types

Total invertebrate abundance did not differ significantly between NF and INT1 S or INT1 P rewetted sediments in 2006 (2-way ANOVA with depth $\times$ riparian vegetation type: $F_{\text{riparian veg}} = 1.10$, $df = 2,30$, $p = 0.35$) or 2007 (0–5 cm, $t = 0.65$, $df = 8$, $p = 0.53$). However, total taxon richness (all taxa recorded/12.5 L sediment over the sampling period) was significantly higher in rewetted sediments from NF and INT1 S than from INT1 P in 2006 (2-way ANOVA with depth $\times$ riparian vegetation, $F_{\text{riparian veg}} = 3.65$, $df = 2,30$, $p = 0.041$) and 2007 (5–10 cm, $t = 4.04$, $df = 8$, $p = 0.004$) (Fig. 5). Sediment depth did not

significantly alter the effect of riparian vegetation on richness in 2006 (ANOVA depth $\times$ riparian vegetation interaction, $F = 0.49$, $df = 2,30$, $p = 0.62$). Invertebrate assemblage composition was significantly different in rewetted sediments from NF than from INT1 S or INT1 P at 0 to 5 cm (2006 and 2007 data combined; ANOSIM global $R = 0.16$, $p = 0.02$; pairwise comparisons of NF vs INT1 P: $R = 0.26$, $p = 0.004$; NF vs INT1 S, $R = 0.10$, $p = 0.16$; INT1 P vs INT1 S, $R = 0.00$, $p = 0.42$) for sediments collected at 0 to 5 cm but not for sediments collected at 5 to 10 cm (2006 data only; ANOSIM global $R = 0.20$, $p = 0.06$). The main distinctions between NF and INT1 S/INT1 P communities at 0 to 5 cm were higher proportions of Oligochaeta and Diptera (Chironomidae and other families) and lower proportions of microcrustacea, Acari, and Gastropoda in NF communities.

Temperature tolerances of over-summering invertebrates in rewetted sediments

In 2006, sediments exposed to cool (10–18°C) temperatures prior to rewetting contained 25 to 430 individuals/12.5 L sediment (averaged by site), whereas sediments exposed to warm (12–30°C) temperatures contained 1.5 to 355 individuals/12.5 L sediment (Fig. 6). Total invertebrate abundance/12.5 L sediment was significantly lower (mean = 43%, range: 11–96%) in the warm- than in the cool-temperature treatment (paired-samples t -test, $t = 3.72$, $df = 15$, $p = 0.002$; Table 4). In the cool treatment, rewetted sediments from NF had higher invertebrate abundances than those from INT1 S or INT1 P, but the % reduction in warm-treatment rewetted sediments (defined as the difference in invertebrate abundance between warm and cool treatments as a percentage of the cool-treatment abundance) was not significantly different among NF, INT1 S and INT1 P sediments (Kruskal–Wallis test, $\chi^2 = 2.27$, $df = 2$, $p = 0.32$; Table 4). Averaged across all tubs, the abundance of each major taxon was between 11 and 94% lower in warm- than in cool-treatment rewetted sediments. Taxa with the lowest % reduction were Ostracoda, Harpacticoida and Cyclopoida (Copepoda), *Gyraulus* (Gastropoda), Naididae (Oligochaeta), and Nematoda. Taxon richness was 5 to 11.3 taxa/12.5 L sediment (averaged by site) in the cool treatment and 1.5 to 11 taxa/12.5 L sediment (averaged by site) in the warm treatment. On average, taxon richness was 1.5 taxa/12.5 L sediment (range 0 to 3.5 taxa/12.5 L sediment) higher in the cool than in the warm treatment (paired-samples t -test, $t = 2.74$, $df = 15$, $p = 0.015$; Table 4). However, very few taxa (only

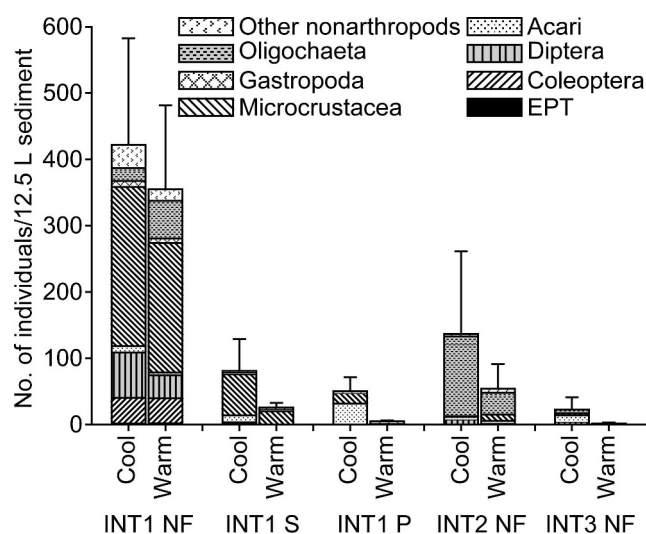


FIG. 6. Mean (± 1 SE of total abundance) abundances of active invertebrates in rewetted streambed sediment samples after incubation of dry sediments in cool- (10–18°C) and warm- (12–30°C) temperature incubators in 2006. EPT = Ephemeroptera, Plecoptera, and Trichoptera, INT = intermittent stream, NF = native forest, S = shaded pasture, P = open pasture. $n = 4, 4, 4, 2$, and 2 for INT1 NF, INT1 S, INT1 P, INT2 NF, and INT3 NF, respectively.

Dytiscidae, Stratiomyidae, and *Physa* (Gastropoda) were absent from all warm-treatment tubs. In both the cool and warm treatments NF had higher taxon richness than INT1 P or INT1 S, but the % reduction in richness between cool and warm treatments did not differ significantly among NF, INT1 S, and INT1 P (Kruskal-Wallis test, $\chi^2 = 1.96$, $df = 2$, $p = 0.38$; Table 4).

TABLE 4. Mean (± 1 SE) differences in total invertebrate abundance and taxon richness in rewetted dry sediment samples collected at 2 depths (0–5 and 5–10 cm) and incubated for 6 wk at warm or cool temperatures before rewetting. Differences are expressed as pairwise comparisons between the 2 treatments/sample, averaged by site. Positive values represent greater numbers in the cool-temperature treatment. “% difference” means the difference between the warm-temperature and the cool-temperature value expressed as a percentage of the cool-temperature value. n.d. = not determined. INT1–3 = intermittent stream sites 1 to 3, NF = native forest, S = shaded pasture, P = open pasture.

Year	Site	<i>n</i>	Difference in total abundance	% difference in total abundance	Difference in total richness	% difference in total richness
2006	INT1 NF	4	67 $\pm$ 50	23 $\pm$ 12	0.3 $\pm$ 1.5	3 $\pm$ 12
	INT1 S	4	55 $\pm$ 45	11 $\pm$ 49	1.5 $\pm$ 1	24 $\pm$ 14
	INT1 P	4	46 $\pm$ 21	83 $\pm$ 7	2.3 $\pm$ 0.3	48 $\pm$ 5
	INT2 NF	2	83 $\pm$ 88	12 $\pm$ 53	0 $\pm$ 1	2 $\pm$ 15
	INT3 NF	2	21 $\pm$ 17	96 $\pm$ 4	3.5 $\pm$ 1.5	81 $\pm$ 19
	All NF samples	8	59 $\pm$ 30	39 $\pm$ 17	1 $\pm$ 0.9	22 $\pm$ 15
	All samples	16	55 $\pm$ 18	43 $\pm$ 15	1.4 $\pm$ 0.5	29 $\pm$ 8
2007	INT1 NF 0–5cm	4	566 $\pm$ 108	96 $\pm$ 1	10 $\pm$ 2.3	72 $\pm$ 9
	INT1 P 0–5cm	4	623 $\pm$ 160	94 $\pm$ 1	2.8 $\pm$ 0.6	44 $\pm$ 7
	INT1 P 5–10cm	4	311 $\pm$ 123	80 $\pm$ 9	3 $\pm$ 1.3	41 $\pm$ 18
	INT2 NF 0–5cm	2	2870 $\pm$ 2330	99.9 $\pm$ 0	10.5 $\pm$ 1.5	88 $\pm$ 13
	All NF samples	6	1334 $\pm$ 776	97 $\pm$ 1	10.2 $\pm$ 1.5	77 $\pm$ 7
	All samples	14	838 $\pm$ 372	91 $\pm$ 3	6 $\pm$ 1.1	58 $\pm$ 7

The 2007 warm treatment (20–37°C) was 7 to 9°C warmer than the 2006 warm treatment. In 2007, cool-treatment (10–18°C) rewetted sediments contained >400 individuals belonging to 6 to 13.3 taxa/12.5 L sediment (averaged by site), whereas warm-treatment rewetted sediments contained ≤ 45 individuals belonging to 1.5 to 3.5 taxa/12.5 L sediment (Fig. 7). The differences in total invertebrate abundance and taxon richness between warm and cool treatments (Table 4) were highly significant (paired samples *t*-test, abundance: $t = 7.43$, $df = 13$, $p < 0.00001$; richness: $t = 4.89$, $df = 13$, $p < 0.001$). Percent reduction in invertebrate abundance in the warm- relative to the cool-treatment tubs did not differ significantly between NF and INT1 P (Mann-Whitney $U = 4.0$, $n = 6,4$, $p = 0.11$; Table 4), but loss of taxon richness in the warm relative to in the cool treatment was significantly greater for NF (mean loss = 10.2 taxa/12.5 L sediment) than INT1 P 0–5 cm (mean loss = 2.8 taxa/12.5 L sediment; Mann Whitney $U = 0.5$, $n = 6,4$, $p = 0.009$; Table 4). Abundances of all major taxa were $\geq 75\%$ lower in warm- than in cool-treatment tubs for all sites, except that Acari and an unidentified beetle larva were more abundant in rewetted INT1 P sediments in warm- than in cool-treatment tubs.

In INT1 P, invertebrate abundance did not differ between surface and subsurface rewetted sediments in the 2007 cool-temperature treatment (paired samples *t*-test, $t = 1.11$, $df = 6$, $p = 0.31$; Fig. 7). Differences in invertebrate abundance between the cool- and warm-temperature treatments were at least as great in the surface as in the subsurface samples. Mean taxon richness/12.5 L sediment sample was

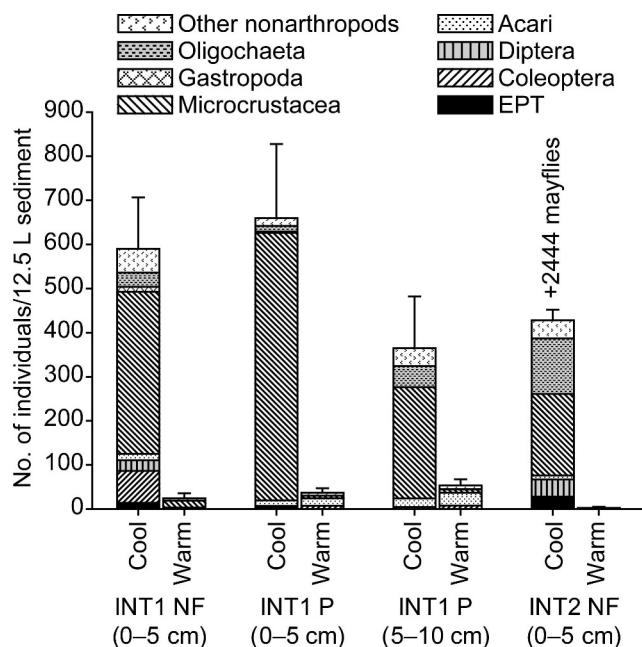


FIG. 7. Mean (+1 SE) abundances of active invertebrates in rewetted surface (0–5 cm) and subsurface (5–10 cm) streambed sediment samples after incubation of dry sediments in cool- (10–18°C) and warm- (20–37°C) temperature incubators in 2007. EPT = Ephemeroptera, Plecoptera, and Trichoptera, INT = intermittent stream, NF = native forest, P = open pasture. $n = 4, 4, 4,$ and 2 for INT1 NF 0–5 cm, INT1 P 0–5 cm, INT1 P 5–10 cm, and INT2 NF 0–5 cm, respectively).

almost equal in surface and subsurface samples (6.0 and 6.5 taxa/12.5 L sediment, respectively) in the cool-temperature treatment and declined to very similar levels (3.25 and 3.5 taxa/12.5 L sediment in surface and subsurface samples, respectively) in the warm-temperature treatment.

Discussion

Our results suggest that benthic invertebrates use dry sediments as refugia in intermittent streams. Of the 53 invertebrate taxa found in the sediments of flowing streams, 37 became active after dry sediment samples collected in 2006 and 2007 were rewetted in the laboratory. The proportion of benthic taxa recovered from rewetted sediments may be an underestimate because fewer dry sediment samples than flowing-stream samples were collected at NF sites. By comparison to the dry sediments, 41 taxa were found in remnant pools, probably the other main dry season refuge, in January 2006 (Storey and Quinn 2008). We have not surveyed other potential refugia exhaustively, but results from a few samples from hyporheic sediments, in-channel leaf packs and

wood, and bankside ground-cover vegetation suggest that these potential refugia did not harbor significant numbers of taxa (RGS, unpublished data). About 30 taxa were found in rockface seepages adjacent to INT2 and INT3, but many of these taxa differed from taxa found in the flowing streams (RGS, personal observations). Drift of individuals from perennial headwaters was not possible in the study streams because these streams dried from their upstream sources to points downstream. Downstream perennial reaches were 3 km, 300 m, and 500 to 800 m away from the INT1, INT2 and INT3 sites, respectively. These downstream perennial reaches could potentially provide colonists to the dry stream reaches via aerial movement of most insect taxa to INT2 and INT3 and possibly of some caddisfly taxa to INT1 (Winterbourn et al. 2007), but significant long-distance, upstream recolonization by flightless invertebrates would be unlikely.

The number of taxa and the proportion of the benthic invertebrate assemblage that we recovered from dry shallow streambed sediments are among the highest values reported. Other investigators have found 27 taxa (<50% of the benthic assemblage), 38 taxa (unknown %), 6 taxa (27% of the benthic assemblage), and 50 taxa (65% of the benthic assemblage) in rehydrated temporary river-bed sediments (Boulton 1989, Stubbington et al. 2009, Chester and Robson 2011, and Datry et al. 2012, respectively). Among the few investigators who compared taxon richness among different refugia, most found many more taxa in remnant pools than in streambed sediments (15 vs 10 taxa, Williams and Hynes 1977; 68 vs 27 taxa, Boulton 1989; 16 vs 6 taxa, Chester and Robson 2011). In contrast, our data suggest that shallow sediments may be nearly as important as remnant pools and that they harbor some taxa that do not appear to survive in remnant pools (e.g., harpacticoid copepods and most Plecoptera). Our results also showed that most intermittent-stream taxa could survive in situ during a summer dry period of up to 9 to 12 wk, whereas in some intermittent stream systems, most taxa are thought to recolonize each year from nearby permanent waters (Smith et al. 2003, Clarke et al. 2010). The relatively high proportion of benthic taxa in dry bed sediments at our study sites is probably a result of the shallow water table and dense riparian vegetation, which lead to high sediment moisture content and mild sediment temperatures at NF sites.

The most abundant major taxa in dry bed sediments in our study (microcrustacea, Oligochaeta, and Nematoda) were similar to those in most other comparable studies (Williams and Hynes 1977,

Paltridge et al. 1997, Larned et al. 2007, Stubbington et al. 2009, Datry et al. 2012) and the families encountered were, with the exception of a few families of Trichoptera, Plecoptera, and Coleoptera, similar to those found in 2 Australian streams by Boulton (1989). The life stages of the different taxa we found in dry bed sediments also were similar to those recorded in other countries (Boulton 1989, Williams 2006). Mayflies, stoneflies, most caddisflies, and scirtid beetles had over-summered as eggs or 1st-instar larvae, whereas harpacticoid copepods were mating within days of rewetting, and gastropods appeared at all stages of maturity. This similarity among geographically dispersed studies suggests that animals with common phylogeny and gross morphology respond to the same environmental pressure with similar behavioral and life-history strategies.

Almost all taxa found in dry sediments were active from day 0, and taxon richness changed very little over time in our experiments (though 1 new taxon [Cladocera] did become active on day 21 in 2006). This suggests that few additional taxa would have been recorded had we run our experiments longer. Abundances of some taxa (ostracods, copepods, and *Potamopyrgus*) were highly variable over the 21 d, probably because of failure to dislodge invertebrates during sampling on later dates. Neither mortality nor predation were likely explanations because <8 dead individuals were found among all tubs on any 1 sampling date, and predators occurred in very low numbers. Few studies exist with which to compare the temporal patterns we found. Like us, Boulton (1989) reported that microcrustaceans became active within hours of inundation of river sediments, whereas Jenkins et al. (2002) found that macroinvertebrate taxon richness increased for 7 to 14 d after inundation of floodplain lake sediments. Paltridge et al. (1997) found some turnover of taxa during the first 14 d after inundation. In our study, invertebrate density increased significantly 14 to 21 d after rewetting stream sediments. In contrast, Tronstad et al. (2005) found that densities of aquatic invertebrates (measured biweekly) in floodplain soils did not change significantly over 10 wk of inundation.

We were surprised that in our study so many taxa became active on day 0 because diapause and delayed egg hatching are common responses to drought (Service 1977, Williams 2006). Some staggered egg hatching probably occurred because invertebrate abundance increased between days 8 and 21, and unhatched eggs (mean = 3/12.5 L sediment, a value higher than the density of Ephemeroptera, Plecoptera, Trichoptera [EPT] taxa but lower than the density of microcrustacea and other taxa) were found at the end

of the 21 d experiment. The effect on the invertebrate community of a false start to the flow period would be worth investigating in future work.

Temperature tolerances of over-summering aquatic invertebrates

In the 2006 and 2007 experiments, the warm temperature treatment caused major declines in invertebrate abundances and significant loss of taxon richness. Maximum temperatures in the middle of each dry sediment sample (estimated at 22–25°C in 2006 and measured at 31–32°C in 2007) were similar to the upper temperature tolerances of the active aquatic stages of several common New Zealand taxa. For example, Quinn et al. (1994) reported 50% mortality of several aquatic invertebrate taxa after 96 h of exposure to water temperatures between 22.6 and 32.6°C. Therefore, among the taxa observed in our study, temperature tolerances appeared to differ little between over-summering and active aquatic stages, as Brittain (1977), Friesen et al. (1979), and Juliano et al. (2002) also found. This similarity in thermal resistance is in contrast with desiccation resistance, which is thought to be higher in over-summering than in active aquatic stages of many taxa (Boulton 1989, Williams 2006). Quinn et al. (1994) reported that the most temperature-sensitive taxa were 2 mayflies and a stonefly, the most tolerant were a riffle beetle and a gastropod, whereas a caddisfly, 2 crustaceans, and an oligochaete showed moderate thermal tolerance. In our experiments, ostracods, copepods, a gastropod, naids (Oligochaeta), and nematodes suffered much higher % reduction in the 31 to 32°C treatment (2007 warm treatment) than in the 22 to 25°C treatment (2006 warm treatment). EPT taxa suffered 100% reduction in the 31 to 32°C treatment, but reduction in the 22 to 25°C treatment could not be estimated with confidence because abundances were low. Therefore, as far as we could assess, temperature tolerances of taxa in their over-summering stages were correlated with those in their active stages. Moreover, % reduction was consistently and significantly higher in the 2007 warm treatment (31–32°C) than in the 2006 warm treatment (22–25°C).

Effects of riparian vegetation on invertebrate survivorship and temperature tolerances

Temperature has been implicated as a key mechanism mediating the correlation between riparian shading and invertebrate abundance or composition in perennial streams (Quinn et al. 1997, Rutherford et al. 1997, Parkyn et al. 2003, Sponseller et al. 2001). Therefore, we expected that the temperature to which

dry sediments were exposed in the field would influence invertebrate abundances and richness in rewetted sediments. We anticipated lower abundance and richness in rewetted sediments from unshaded, exposed INT1 P sites than in sediments from forested NF sites, but we expected invertebrates from INT1 P sites to have lower mortality in the warm treatment. Taxon richness and community composition did differ among INT1 P, NF, and INT1 S sites, but our study design did not permit us to rule out all factors except temperature as the underlying mechanism for these results. Percent reduction of taxa and of individuals in the 2007 warm treatment was higher in NF than in INT1 P tubs, but lower in NF and INT1 S than in INT1 P tubs in the 2006 warm treatment. Therefore, we cannot conclude from our laboratory experiments that taxa over-summering in the INT1 P site are more tolerant of high temperatures than those over-summering in NF or INT1 S sites. We found only 1 study in an intermittent stream with which to compare these results. Aguiar et al. (2002) found that cover (shade) by riparian vegetation was a strong predictor of variation in invertebrate community composition among intermittent streams in Portugal, but they attributed this result primarily to the effect of riparian trees on the food base rather than on temperature.

Among the INT1 P samples, we expected lower abundance and richness in rewetted surface sediments than in subsurface sediments because surface sediments are naturally exposed to greater solar heating than are subsurface sediments. We further expected that % reduction in the warm-temperature treatments would be lower for invertebrates in surface than in subsurface rewetted sediments. Our data did not support either of these predictions. However, this result does not necessarily disprove the importance of temperature in structuring the invertebrate community. Invertebrate abundances might have been low in surface sediments because temperatures were $>35^{\circ}\text{C}$, and low in subsurface sediments for some other reason, such as siltation or low O_2 during the flow period (Boulton et al. 1997). In the thermal tolerance experiment, % reduction was so high in the warm-temperature treatment that it may have obscured any differences in thermal tolerance among taxa occurring in surface vs subsurface samples.

Our field measurements indicate that intact riparian forest can reduce daily temperature maxima in dry stream beds from levels that were associated with high % reduction in laboratory experiments to levels at which most taxa survived (as shown for flowing waters by Rutherford et al. 1997). Therefore, protecting or restoring riparian forests may be critical for protecting the refugia that allow aquatic invertebrates

to survive summer drought in intermittent streams, just as it is critical for maintaining invertebrate communities of perennial streams (Storey and Cowley 1997, Parkyn et al. 2003). Many streams in New Zealand are shaded by scattered tree cover rather than intact riparian forest. Below the scattered tree cover at our study site, temperatures in the top 5 cm of sediment reached daily maxima of 22 to 29°C (similar to temperatures in the 2006 warm-temperature treatment) compared to 35 to 42°C in open pasture (a little warmer than the 2007 warm-temperature treatment). Percent reduction in the warm-temperature treatment was consistently lower in the 2006 than in the 2007 experiment (11–96% reduction compared to 94–99.9% reduction in invertebrate abundance relative to the cool-temperature treatment in NF sediments). Thus, it appears that scattered tree cover giving $\geq 50\%$ shading may partially protect dry-sediment refugia from high temperatures, but intact riparian forest would afford greater protection. This conclusion is consistent with Rutherford et al. (1997), whose model showed that in perennial streams, 50% shading was sufficient to maintain water temperatures within invertebrate tolerance limits, and that 70% shading further reduced daily maximum water temperatures by 5°C .

Conclusions

The resilience of invertebrate communities after stream drying depends on the existence of viable drought refugia (Boulton and Lake 2008), and the importance of protecting refugia to maintain biodiversity and ecological functioning of intermittent streams has been highlighted by many authors (e.g., Sedell et al. 1990, Bond et al. 2008, Boulton and Lake 2008). Most investigators recommend protecting habitat quality and connectivity of perennial pools (Sheldon et al. 2010, Chester and Robson 2011) and nearby perennial streams (Chester and Robson 2011), maintaining the flow and sediment structure of hyporheic zones (Dole-Olivier 2011, Stubbington 2012), and protecting shallow bed sediments from effects of recreation, gravel abstraction, and pollution (Datry et al. 2012). We conclude that shallow stream bed sediments can be a key dry-season refuge for invertebrates in intermittent streams and that protecting or restoring riparian vegetation is critical for protecting the refuge potential of this habitat by moderating summer streambed temperatures.

Acknowledgements

We thank Brian Smith for his expert advice and help with identifications; Carola and Mike Hudson,

Spencer and Gordon MacDonald for help with sample collection and their observations of local flow patterns; Graham Sevicke-Jones, Brett Stansfield, Vickie Hansen, and Andrew Lamason for help with sample collection; and Holly Ferguson, Claire Taylor, and Nathan Wagstaff for help with sample sorting. This research was funded by the Foundation for Research, Science and Technology through a postdoctoral fellowship (NIWX0401) to RGS.

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Received: 18 January 2012
Accepted: 2 November 2012